

ETF5231 Group Assignment 3

Group7 GenieUS

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For the tasks that follow you will be modelling the original data without any transformation performed (even if you previously deemed it necessary).

Question 1

Plot your time series. By observing the plot and describing its components select an ETS model you think is appropriate for forecasting. Make sure you justify your choice (no more than 150 words). (8 marks)

```
print_retail_act |>
  autoplot(y) +
  labs(x = "Month",
       y = "Turnover (Millions of Dollars)",
       title = "Newspaper and book retailing",
       subtitle = "Australian Capital Territory")
```

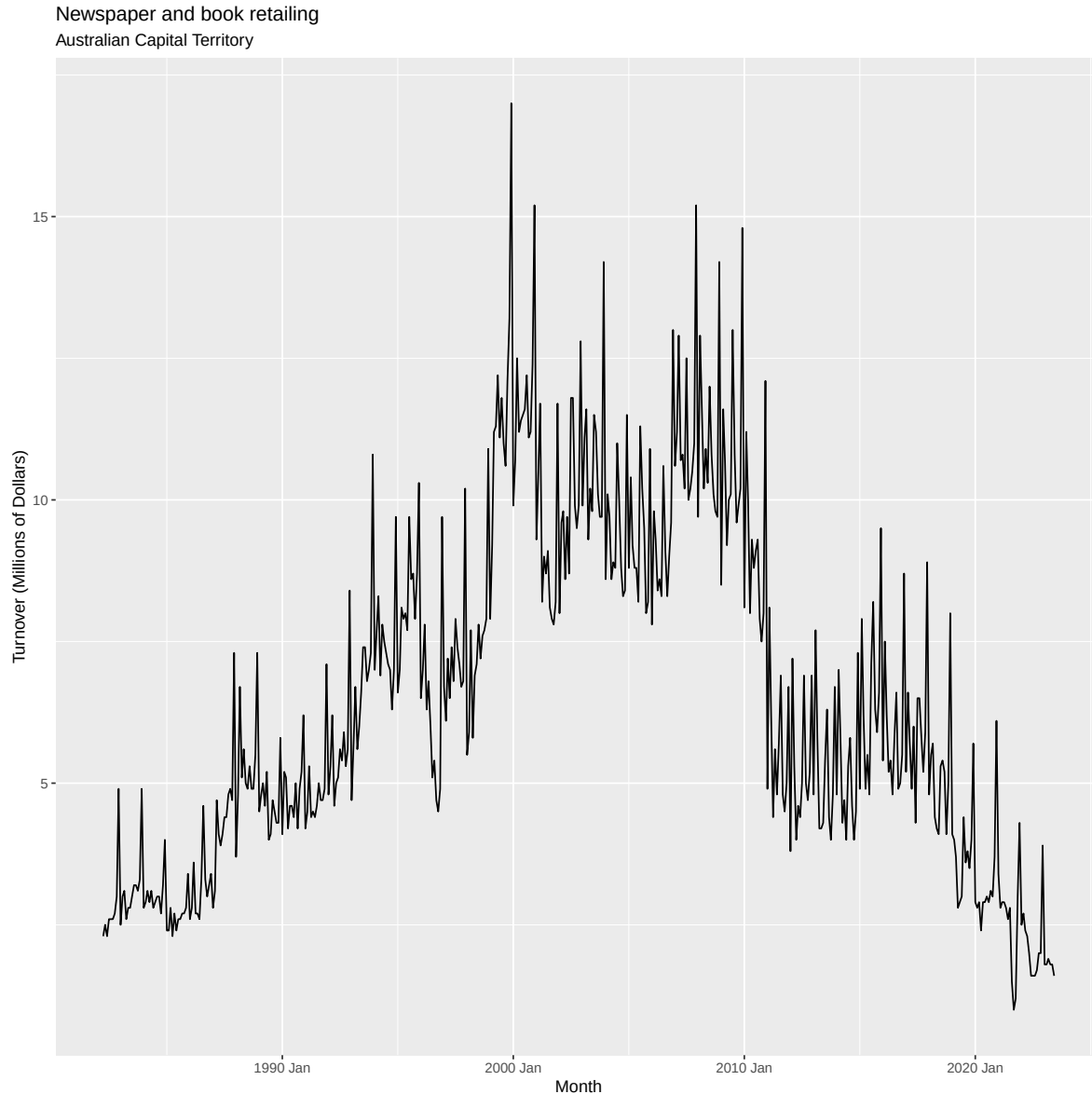


Figure 1: Time plot of monthly newspaper and book retail turnover in ACT (1982-2023).

As shown in Figure 1, the ACT newspaper and book retailing series has clear monthly fluctuations and a changing overall level. The series rises in the earlier years, stays relatively high around the middle of the sample, and then falls strongly in the later years. This suggests that there is no stable long-term trend, but there are strong seasonal effects.

Because the direction of the series changes over time, we do not think an additive or damped trend which assumes a consistent trend direction is suitable. The size of the seasonal fluctuations

and the random variation also seem larger when turnover is high, and smaller when turnover is low. This means the errors are more likely to be proportional to the level, so a multiplicative error structure is reasonable.

Therefore, we choose an **ETS(M,N,M)** model, which allows multiplicative errors and seasonality without forcing a fixed trend.

Question 2

Search for an ETS model for your series using appropriate information criteria. Describe the process you have implemented. Which part of the data is used in this process? Which model have you selected and why? Show all necessary output (10 marks)

```
auto_fit <- print_retail_act |>
  model(auto_ets = ETS(y))

report(auto_fit)
```

Series: y

Model: ETS(M,N,M)

Smoothing parameters:

alpha = 0.4017636

gamma = 0.3184233

Initial states:

l[0]	s[0]	s[-1]	s[-2]	s[-3]	s[-4]	s[-5]	s[-6]
2.667772	1.10443	0.9561735	0.8708791	1.505046	1.001178	0.9256466	0.9452733
s[-7]	s[-8]	s[-9]	s[-10]	s[-11]			
1.003178	0.9604205	0.8919527	0.9531788	0.8826433			

sigma^2: 0.015

AIC	AICc	BIC
2744.476	2745.478	2807.544

```
auto_fit |>
  components() |>
  autoplot()
```

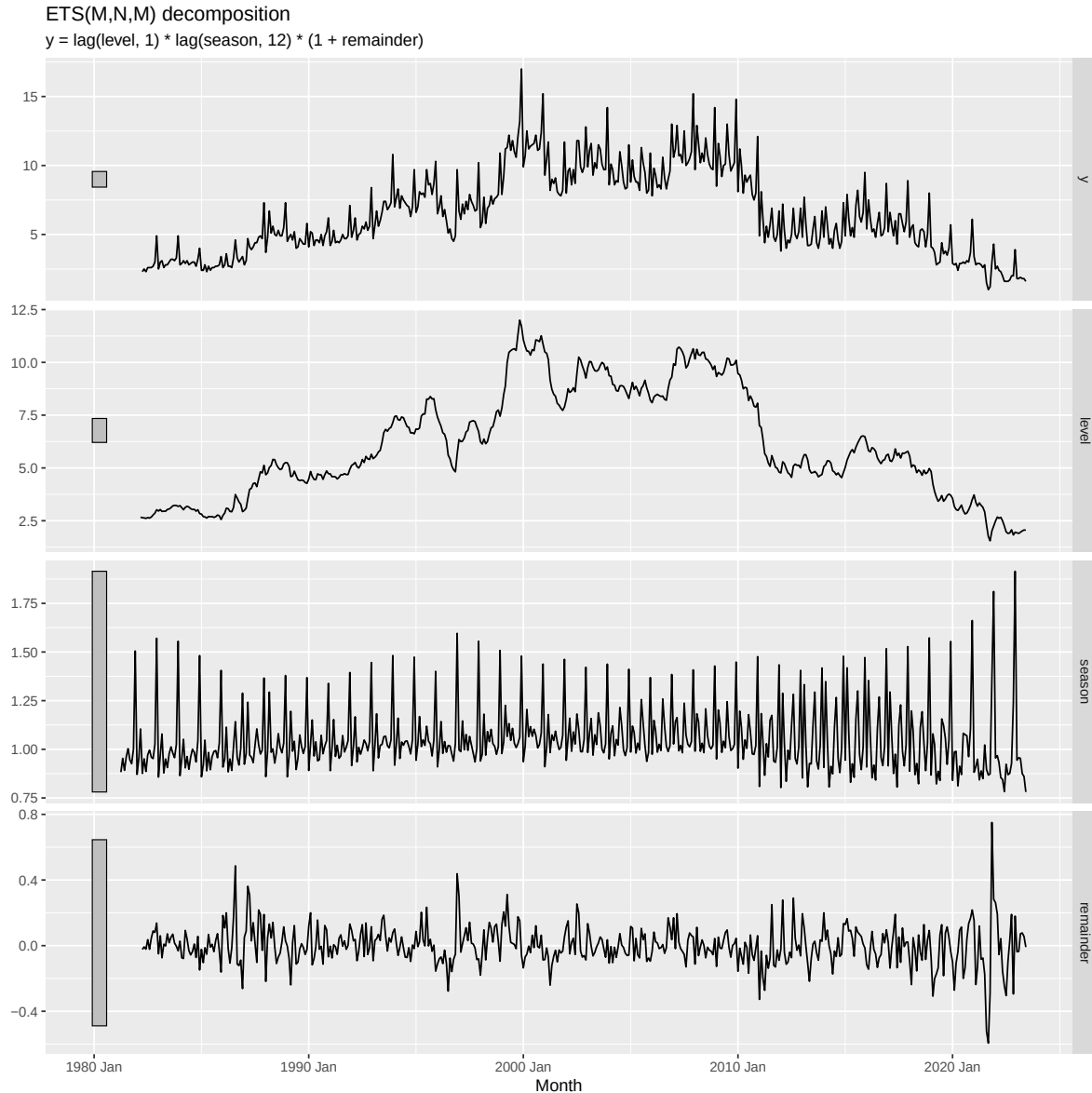


Figure 2: Estimated components from the automatically selected ETS(M,N,M) model.

The `ETS()` function considers the admissible ETS models and selects the model with the lowest AICc, balancing goodness of fit and model complexity. The automatic search selected **ETS(M,N,M)**, which is consistent with our choice in Question 1. The entire available time series (1982–2023) was used in this process, because this question focuses on information criteria rather than out-of-sample evaluation.

The output from `report()` shows that the selected model has multiplicative errors, no trend

and multiplicative seasonality. The reported information criteria are $AIC = 2744.476$, $AIC_c = 2745.478$ and $BIC = 2807.544$. This supports our visual choice, because both the time plot and the AICc-based search suggest proportional errors and seasonal variation, but no stable long-term trend.

Question 3

Now use time series cross-validation to compare the model selected from the previous step, along with $ETS(M,N,M)$, $ETS(M,A,M)$ and $ETS(M,Ad,M)$ models. You will need to think about more than $h=1$ step-ahead. Think about an appropriate range of forecast horizons and justify your choice. Fully describe the process you have implemented. Show all necessary output. (Hint the code at the end of this section <https://otexts.com/fpp3/tscv.html> may help you to get started.) (22 marks)

```
print_retail_cv <- print_retail_act |>
  stretch_tsibble(.init = 180, .step = 1)

cv_fit <- print_retail_cv |>
  model(mnm = ETS(y ~ error("M") + trend("N") + season("M")),
        mam = ETS(y ~ error("M") + trend("A") + season("M")),
        madm = ETS(y ~ error("M") + trend("Ad") + season("M")))

cv_fc_12 <- cv_fit |>
  forecast(h = 12)

cv_acc_12 <- cv_fc_12 |>
  accuracy(print_retail_act) |>
  arrange(RMSE)

cv_acc_12 |>
  select(.model, .type, RMSE, MAE, MAPE, MASE, RMSSE, ACF1) |>
  kable(digits = 3)
```

Table 1: Time series cross-validation accuracy for the candidate ETS models using a 12-month forecast horizon.

.model	.type	RMSE	MAE	MAPE	MASE	RMSSE	ACF1
mnm	Test	1.295	0.937	15.638	0.890	0.871	0.739
madm	Test	1.304	0.963	15.915	0.915	0.877	0.712
mam	Test	1.358	1.015	17.262	0.964	0.914	0.724

```
cv_fc_12_h <- cv_fc_12 |>
  group_by(.id, .model) |>
  mutate(h = row_number()) |>
  ungroup() |>
  as_fable(response = "y", distribution = y)
```

```

cv_fc_12_h |>
  accuracy(print_retail_act, by = c("h", ".model")) |>
  ggplot(aes(x = h, y = RMSE, colour = .model)) +
  geom_point() +
  geom_line() +
  scale_x_continuous(breaks = seq(1, 12, by = 2)) +
  labs(x = "Forecast horizon", y = "RMSE",
       colour = "Model",
       title = "Cross-validation RMSE by forecast horizon",
       subtitle = "12-month forecast horizon") +
  theme_minimal()

```

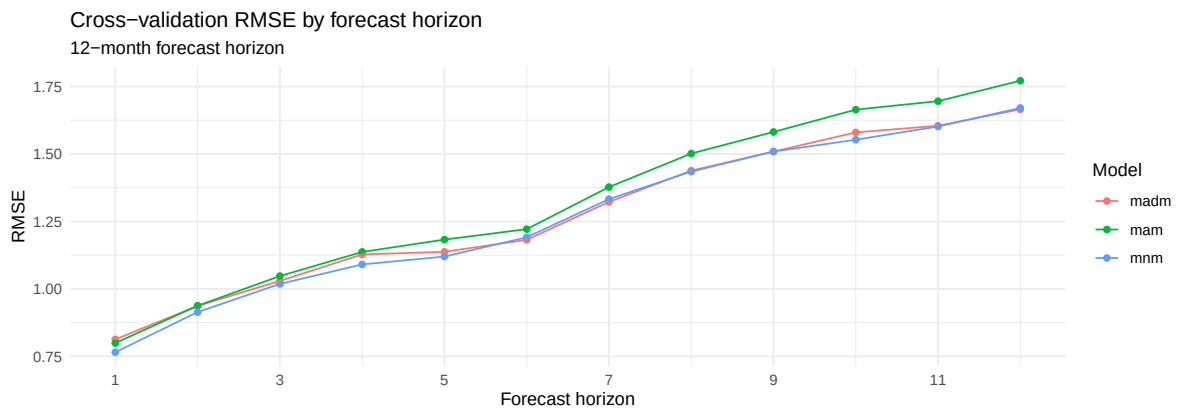


Figure 3: Cross-validation RMSE by forecast horizon for the candidate ETS models using a 12-month forecast horizon.

```

cv_fc_24 <- cv_fit |>
  forecast(h = 24)

cv_acc_24 <- cv_fc_24 |>
  accuracy(print_retail_act) |>
  arrange(RMSE)

cv_acc_24 |>
  select(.model, .type, RMSE, MAE, MAPE, MASE, RMSSE, ACF1) |>
  kable(digits = 3)

```


Table 2: Time series cross-validation accuracy for the candidate ETS models using a 24-month forecast horizon.

.model	.type	RMSE	MAE	MAPE	MASE	RMSSE	ACF1
mnm	Test	1.760	1.267	21.328	1.203	1.184	0.836
madm	Test	1.776	1.297	21.621	1.232	1.194	0.823
mam	Test	1.924	1.427	24.829	1.355	1.294	0.826

```
cv_fc_24_h <- cv_fc_24 |>
  group_by(.id, .model) |>
  mutate(h = row_number()) |>
  ungroup() |>
  as_fable(response = "y", distribution = y)

cv_fc_24_h |>
  accuracy(print_retail_act, by = c("h", ".model")) |>
  ggplot(aes(x = h, y = RMSE, colour = .model)) +
  geom_point() +
  geom_line() +
  scale_x_continuous(breaks = seq(1, 24, by = 2)) +
  labs(x = "Forecast horizon", y = "RMSE",
       colour = "Model",
       title = "Cross-validation RMSE by forecast horizon",
       subtitle = "24-month forecast horizon") +
  theme_minimal()
```

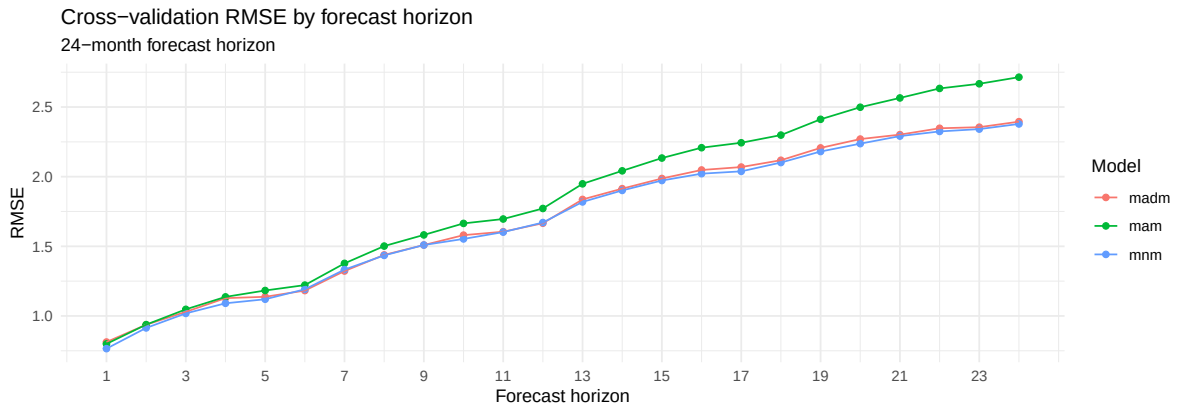


Figure 4: Cross-validation RMSE by forecast horizon for the candidate ETS models using a 24-month forecast horizon.

Time Series Cross-Validation Process:

Time series cross-validation was used to compare the candidate ETS models beyond one-step-ahead forecasts. We used 12-month and 24-month forecast horizons, representing one and two full seasonal cycles. For both horizons, `.init = 180` was used, giving 15 years of initial monthly data, enough to estimate the monthly seasonal pattern. In both cases, `.step = 1` was used so every possible monthly forecast origin after the initial training period was used.

Results:

The 12-month results in Table 1 show that ETS(M,N,M) has the lowest RMSE, MAE, MAPE, MASE and RMSSE, although ETS(M,Ad,M) is very close. In Figure 3, RMSE increases as the forecast horizon increases. The three models are close at early horizons, but ETS(M,A,M) becomes worse at longer horizons, while ETS(M,N,M) and ETS(M,Ad,M) remain similar.

The 24-month results in Table 2 show the same ranking, with ETS(M,N,M) still slightly better overall. In Figure 4, RMSE again increases with horizon. ETS(M,A,M) moves further away from the other two models, while ETS(M,N,M) and ETS(M,Ad,M) almost overlap for much of the horizon.

The close performance of ETS(M,Ad,M) may be because some rolling training windows include the middle part of the series, where the level rises and then begins to flatten or decline. A damped trend can capture this kind of local movement, but the trend is not stable across the whole series, so it does not clearly improve accuracy over ETS(M,N,M).

Conclusion:

The cross-validation results support ETS(M,N,M) as the preferred model. It performs slightly best forecast accuracy across both horizons, so the simpler no-trend model is preferred over the additive or damped trend alternatives.

Question 4

Which model do you prefer for your data? Choose a model and briefly explain your choice. Plot and describe its components and comment on the estimated parameters and how these are appropriate for your data. Perform all the necessary diagnostic checks and comment. (max 400 words) (14 marks)

```
fit_ets_mnm <- print_retail_act |>
  model(ETS_MNM = ETS(y ~ error("M") + trend("N") + season("M")))

report(fit_ets_mnm)
```

Series: y

Model: ETS(M,N,M)

Smoothing parameters:

alpha = 0.4017636

gamma = 0.3184233

Initial states:

l[0]	s[0]	s[-1]	s[-2]	s[-3]	s[-4]	s[-5]	s[-6]
2.667772	1.10443	0.9561735	0.8708791	1.505046	1.001178	0.9256466	0.9452733
s[-7]	s[-8]	s[-9]	s[-10]	s[-11]			
1.003178	0.9604205	0.8919527	0.9531788	0.8826433			

sigma^2: 0.015

AIC	AICc	BIC
2744.476	2745.478	2807.544

```
fit_ets_mnm |>
  components() |>
  autoplot() +
  labs(title = "Components of ETS(M,N,M)",
        subtitle = "Multiplicative error, no trend, and multiplicative seasonality") +
  theme_minimal()
```

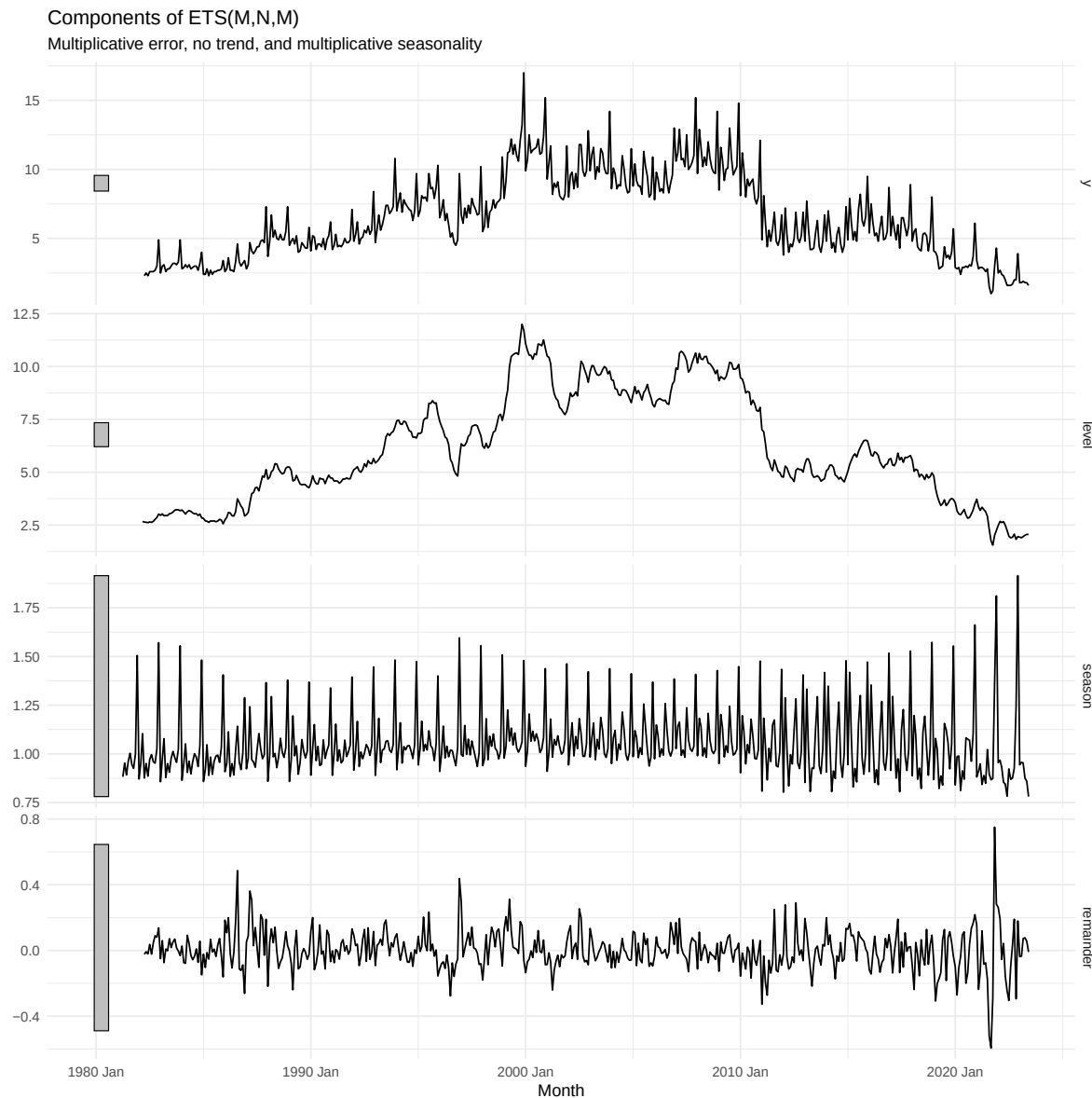


Figure 5: Estimated components from the selected ETS(M,N,M) model.

```
fit_ets_mnm |>
  gg_tsresiduals()
```

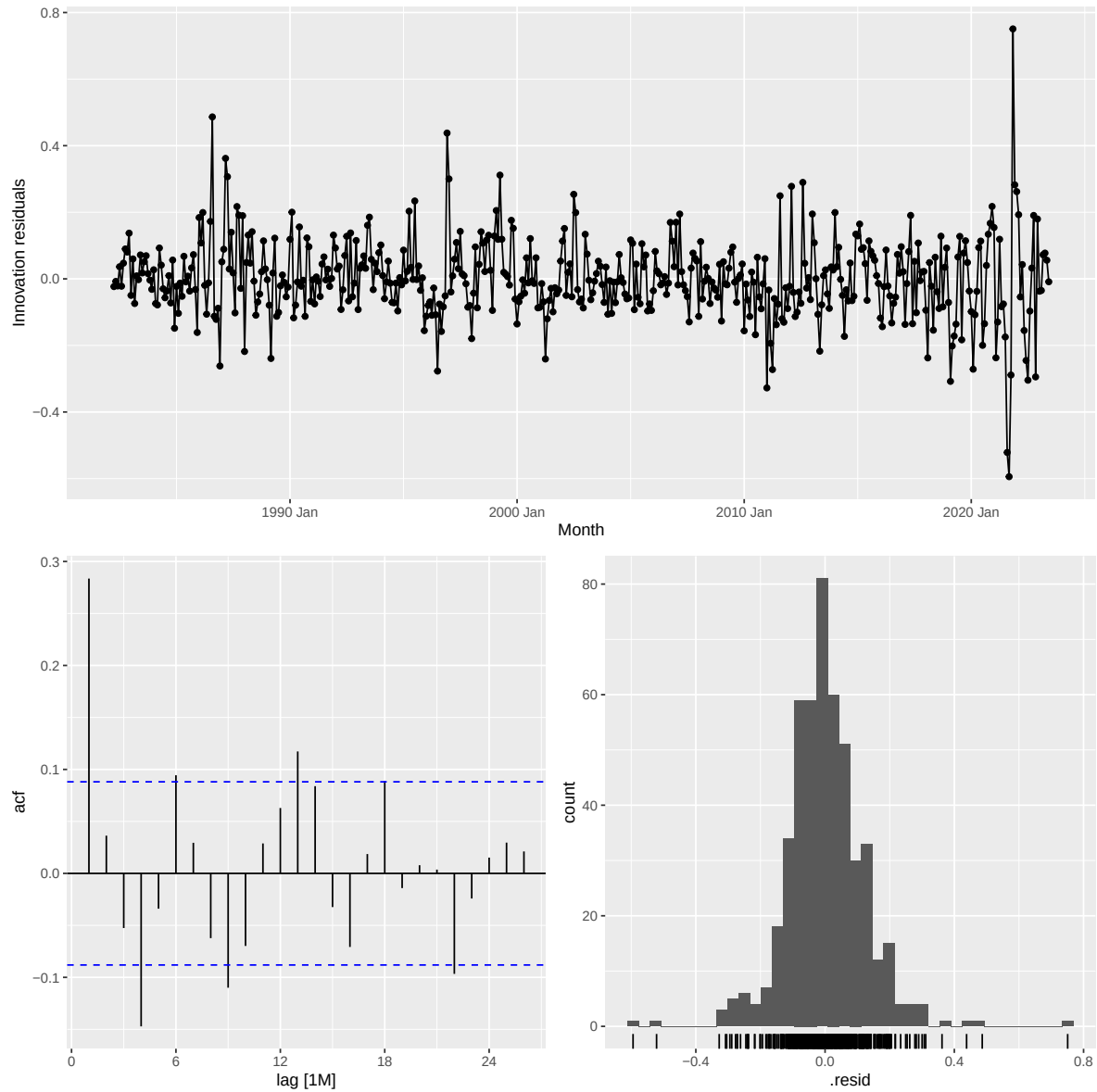


Figure 6: Residual diagnostics for the ETS(M,N,M) model.

```
ljung_fit <- fit_ets_mnm |>
  augment() |>
  features(.innov, ljung_box, lag = 24)

ljung_fit |>
  mutate(lb_pvalue = format(lb_pvalue, scientific = TRUE, digits = 6)) |>
```

```
kable(caption = "Ljung-Box test results for the ETS(M,N,M) model residuals.")
```

Table 3: Ljung-Box test results for the ETS(M,N,M) model residuals.

.model	lb_stat	lb_pvalue
ETS_MNM	94.81622	2.26692e-10

The ETS(M,N,M) model is considered the preferred model because it has the lowest AICc among the ETS candidates and was also supported by the time series cross-validation results. This model allows for multiplicative errors and multiplicative seasonality, with no trend component. This is appropriate because the series shows strong seasonal variation, but no stable linear trend over the full sample.

The estimated smoothing parameters are $\alpha = 0.402$ and $\gamma = 0.318$. These values suggest that both the level and seasonal components are updated moderately as new observations become available. The absence of a trend smoothing parameter is consistent with the ETS(M,N,M) specification.

Figure 5 supports this model choice. The level component changes over time but does not follow a stable linear upward or downward trend, supporting the no-trend specification. The seasonal component shows a strong pattern, which supports the use of multiplicative seasonality. The seasonal fluctuations are consistently present across the sample with varying sizes, especially around the end of the series. The remainder is mostly centred around zero, but some large movements remain, especially around 2020.

Figure 6 shows that the innovation residuals mostly fluctuate around zero, but there are clear outliers around 2020 and slightly larger variation near the end of the sample. The ACF plot shows several spikes outside the significance bounds, suggesting that some autocorrelation remains. The residual histogram is centred near zero, although it shows some right skewness and extreme values.

The Ljung-Box test in Table 3 was applied at lag 24. For ETS(M,N,M), there are 14 estimated parameters, including two smoothing parameters. The Ljung-Box test gives a p-value below 0.05, so the null hypothesis of no residual autocorrelation is rejected. This means the residuals are not consistent with white noise.

Therefore, ETS(M,N,M) captures the main seasonal structure reasonably well, but the residual diagnostics indicate some remaining unexplained dependence.

Question 5

Generate forecasts for the two years following the end of your sample using your chosen/best model. Plot them and briefly comment on these. (No more than 100 words). (4 marks)

```
fc_ets_mnm <- fit_ets_mnm |>
  forecast(h = "24 months")
```

```
fc_ets_mnm |>
  autoplot(print_retail_act, level = c(80, 95)) +
  labs(title = "Two-year forecasts for newspaper and book retailing",
        subtitle = "Australian Capital Territory",
        x = "Month", y = "Turnover (Millions of Dollars)") +
  theme_minimal()
```

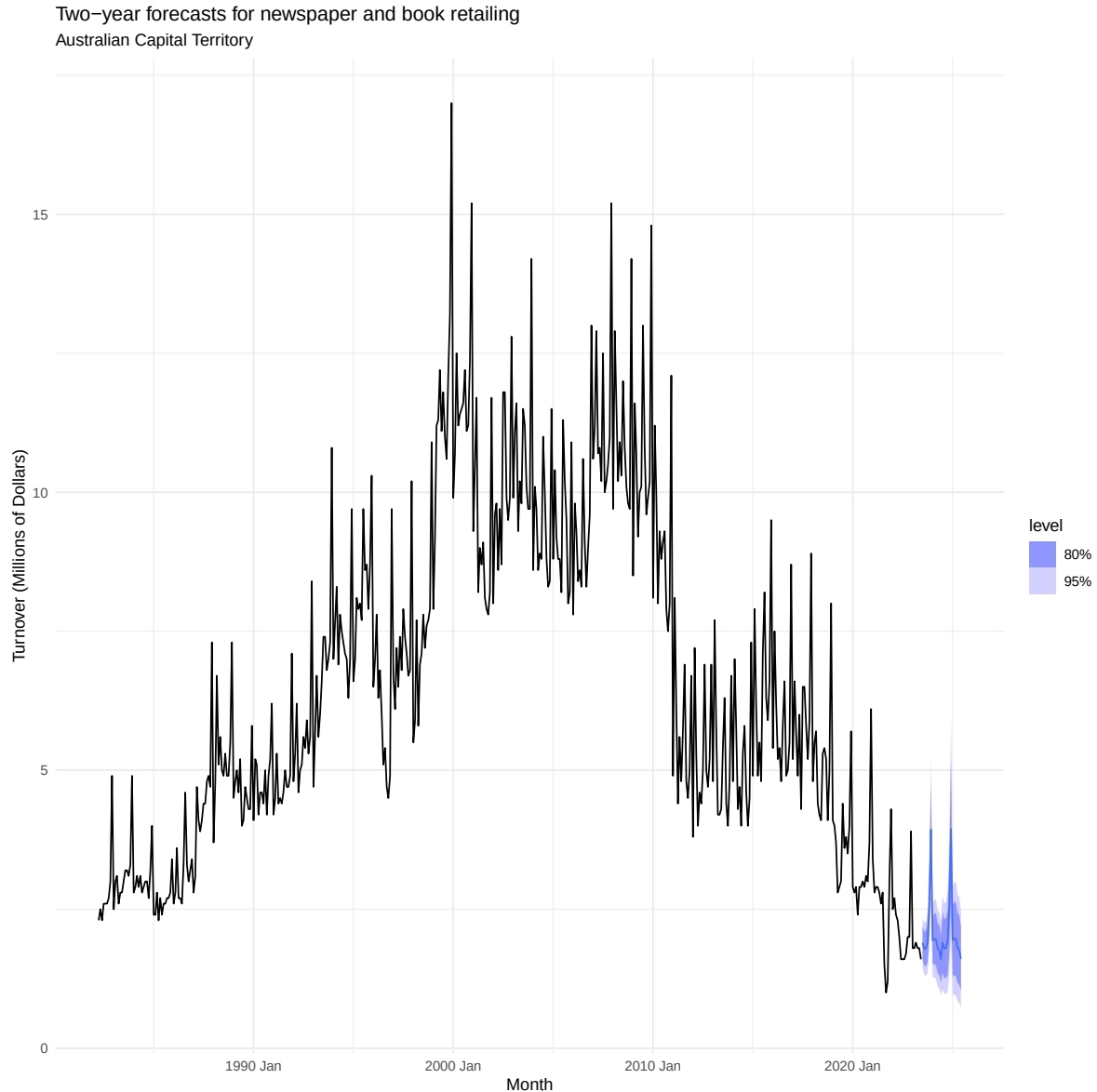


Figure 7: Two-year forecasts for ACT newspaper and book retailing using $ETS(M,N,M)$.

The two-year forecasts continue the pattern supported in Q1–Q4: low turnover, no sustained trend, but clear multiplicative monthly seasonality. The point forecasts remain near recent low levels rather than returning to the high-turnover period around 2000–2010, supporting the no-trend $ETS(M,N,M)$ choice and Q3 cross-validation results. The sharp seasonal peaks indicate monthly effects remain important. The 80% and 95% prediction intervals widen with the forecast horizon, reflecting increasing uncertainty. Since Q4 showed remaining autocorrelation,

the forecast direction and seasonal pattern are useful, but exact values and interval widths may be less reliable.